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Pilot Internet-Quality Comparison by Verified Access Medium

SIGNAL SYNC

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Problem Description

- Traditional speed tests mainly measure peak download bandwidth.
- They may overlook latency, packet loss, and loaded jitter.
- Internet performance varies across fiber, cellular, fixed wireless, and satellite.
- The project follows the RFC 2330 measurement framework.
- It tests 10–20 volunteer endpoints over seven days.
- It measures latency, packet loss, DNS delay, and limited throughput.
- It compares at least two access media, considering location, ISP, Wi-Fi conditions, and measurement uncertainty.
- Results are presented through an interactive comparison dashboard

Background

- Internet connectivity is important for everyday online activities.
- Download speed alone cannot fully describe internet quality.
- Latency, packet loss, DNS delay, and network stability affect performance.
- Different access media can provide different performance results.
- Location, ISP, and local Wi-Fi conditions can influence measurements.
- RFC 2330 provides the measurement framework referenced in the problem statement.
- Scheduled tests over seven days help identify performance variations.
- The collected data is analysed and displayed through an interactive dashboard

Literature & RFC Review

- RFC 2330: Provides the measurement framework referenced in the problem statement for Internet performance metrics.
- Traditional Speed Tests: Mainly measure peak download bandwidth and may overlook other aspects of internet quality.
- Latency Measurement: Measures round-trip delay, which affects the responsiveness of real-time applications.
- Packet Loss Measurement: Identifies packets that fail to reach their destination.
- DNS Resolution Delay: Measures the time required to resolve a domain name.
- Throughput Measurement: Measures the rate of successful data transfer during testing.
- Access-Medium Comparison: The project compares at least two available media, considering location, ISP, Wi-Fi conditions, and measurement uncertainty

Proposed Solution

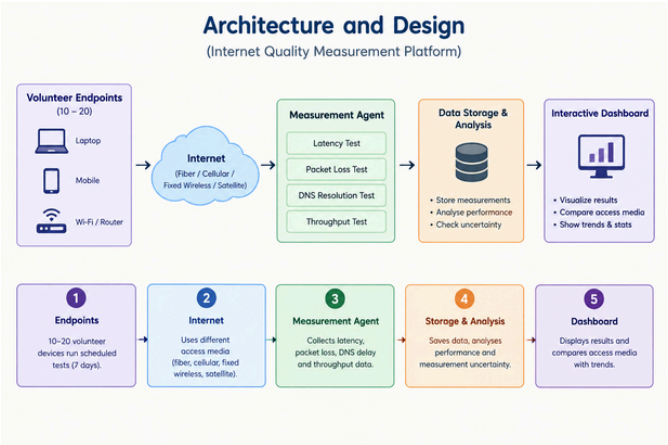
- Measurement Agent: Runs scheduled internet performance tests across volunteer endpoints.
- Controlled Endpoint: Provides a designated destination for consistent network measurements.
- Performance Metrics: Measures latency, packet loss, DNS resolution delay, and limited throughput.
- Access-Medium Comparison: Compares at least two available media, such as fiber, cellular, fixed wireless, or satellite.
- Data Collection and Analysis: Collects data from 10–20 volunteer endpoints over seven days and analyses performance variations and measurement uncertainty.
- Interactive Dashboard: Displays results while considering location, ISP, Wi-Fi conditions, and access-medium information.

Optimization Proposed

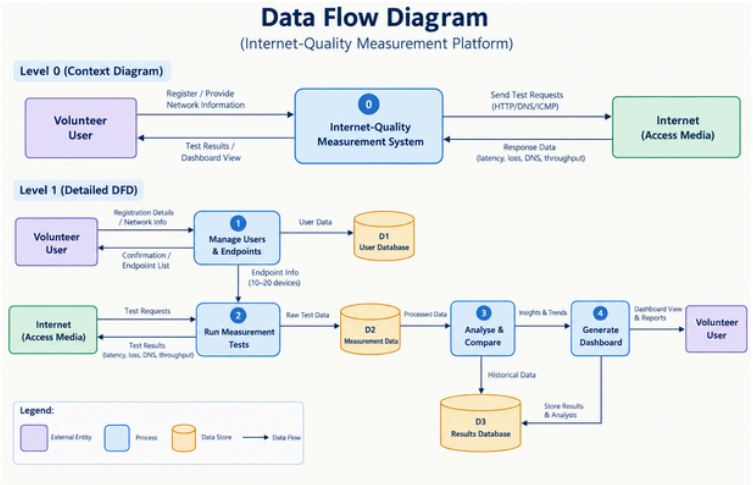
- Consistent Testing: Use the same measurement method during scheduled tests to make results more comparable.
- Repeated Measurements: Collect readings over seven days instead of relying on a single speed test.
- Measurement Accuracy: Identify possible errors and sources of uncertainty in the collected results.
- Clear Data Analysis: Calculate useful summaries and examine changes in network performance over time.
- Fair Comparison: Consider differences in access medium, location, ISP, and Wi-Fi conditions when comparing results.
- Better Presentation: Display results, sample counts, and measurement uncertainty in an interactive dashboard

Before	After
Single or limited reading	Scheduled measurements over seven days
Speed-focused results	Multiple internet-quality metrics
Unclear measurement reliability	Uncertainty and sample-count analysis

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

- Planning: Study the problem statement and RFC 2330, and decide which internet performance metrics to measure.
- Development: Build the measurement agent and controlled measurement endpoint.
- Dashboard Setup: Create a dashboard to display latency, packet loss, DNS delay, and throughput results.
- Volunteer Setup: Select 10–20 volunteer endpoints and record their access-medium information.
- Seven-Day Testing: Run scheduled measurements over seven days and store the collected results.
- Analysis and Demonstration: Compare the measurements, examine uncertainty and performance variations, and prepare the final technical report and demonstration

Bibliography I

[1] Internet Engineering Task Force (IETF). “RFC 2330: Framework for IP Performance Metrics.” May 1998. https://www.rfc-editor.org/rfc/rfc2330.html

[2] AIORI Hackathon. “PS-014: Pilot Internet-Quality Comparison by Verified Access Medium.” Problem Statement PDF, Smart Cities category.

Thank You for Your Attention!